The Star Fort Belt

How the bastion spread around the world:
outlines, dates and sizes of 1,307 star forts from open data

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Abstract

The bastioned fortification, or trace italienne, was devised in Italy around 1500 and built from the Low Countries to Japan over the next three and a half centuries. I assemble an open catalogue of 1,307 star forts in 80 countries: 769 bastioned outlines found by shape in OpenStreetMap, 498 further forts catalogued in Wikidata, and 40 famous forts added from English Wikipedia, after setting aside 253 outlines that are probably towered walls of other traditions and 560 Bavarian listed earthworks of uncertain form. Only 251 forts (19%) can be dated, 84 of them by hand. 89% of the forts lie in Europe, in a belt along the war zones from the Low Countries to the Baltic. The earliest dated works are Italian, but the bastion did not spread outwards from Italy: a country's distance from Florence barely predicts the date of its first dated fort (Spearman's $\rho = 0.15$, 49 countries), and the median first date is 1623 for countries beyond Europe against 1626 within it, because Portuguese and Spanish ships carried it to India, Africa and the Caribbean in the 1530s and 1540s. The traced outlines are small (median 1.9 ha, about 228 m across) and most often have four points (42%); size does not change with date. 59% of the catalogued forts still have an outline that a shape test recognises.

Keywords: star fort; bastion; trace italienne; military architecture; diffusion; OpenStreetMap; Wikidata; historical GIS

1 Introduction

Between the 1490s and the 1530s Italian engineers answered the siege gun with a new kind of wall: low, thick and angled, with projecting bastions placed so that every face could be swept by fire from a neighbour. The resulting star-shaped trace, the trace italienne, became the standard fortification of early modern Europe and of the European empires, and historians have tied it to the growth of armies, the cost of war and the reach of European power overseas (Parker, 1988; Duffy, 1979). Its spread is usually told through famous examples: Michele Sanmicheli's Verona, the ideal city of Palmanova, the Dutch fortress belt, Vauban's France, and the colonial forts from Goa to Havana.

This study asks what an open, systematic catalogue can add. Where are the star forts that survive or are recorded? When were they built, and how far can the dates be trusted? Did the design spread outward from Italy, or did it jump? How large were they, how many points did they have, and how many survive on the ground? The work accompanies an interactive map drawn on The Inked World basemap (Hotham, 2026) at https://maps.tristanhotham.com/star-forts/, with a decade slider and every fort's sources.

2 Data

Outlines. The forts come from The Inked World (Hotham, 2026), which extracted from OpenStreetMap (OpenStreetMap, 2026) every way and relation tagged as a fort or fortification, every town wall and every moat, joined broken walls and moats into rings, and kept the rings that look like a bastioned trace (Section 3). This gives 1,022 outlines. It also added 1,061 forts from Wikidata (Wikidata, 2026) that belong to the classes star fort (Q12797), sconce (Q91203, Q6653030) or polygonal fort (Q92080) but have no traced outline.

Dates and names. For every Wikidata item I queried, with QLever, the inception (P571) with its precision, start time (P580), official opening (P1619), dissolution or demolition (P576), classes and architect, and the full list of items under star fort and fortified town (Q677678) with coordinates and dates. From the OpenStreetMap extract I read the name, `wikidata` and `start_date` tags of every element lying on each outline. Finally, I dated 84 important forts by hand from their English Wikipedia articles (Wikipedia, 2026), recording the year the bastioned works were begun, the builder, and whether the works survive (Appendix, Table 4). Countries and town names come from Natural Earth (Natural Earth, 2026).

3 Methods

3.1 Finding a star

A ring counts as bastioned when, simplified to 4 m, it has a perimeter between 250 m and 20 km, fills between 45% and 93% of its convex hull, and has at least four salient corners: convex corners sharper than 100 degrees that lie further from the centre than their neighbours, at distances from the centre that vary by no more than a quarter, with at least as many re-entrant corners between them. Rings named or catalogued as star forts or sconces need only three. I keep the number of salients as the fort's "points". Because rings joined from broken walls and moats are made with round buffers that soften the bastion points, the maps and Figure 7 draw each fort from its own wall, rampart and moat pieces at full detail instead; the rings are used only to find and measure the forts. Its area, perimeter and largest diameter are measured on a local plane.

3.2 Cleaning the catalogue

Catalogue points lying on an outline were merged into it (3), and Wikidata fortified towns lying inside an outline gave it their name and identifier. Outlines with no name in either source (603) are named after the nearest Natural Earth town. Two groups are kept in the data but left out of every count: 560 entries from the Bavarian monument list that Wikidata labels as sconces, whose form and age are unknown; and 253 outlines in countries where the shape test mostly finds fortifications of other traditions (the towered walls of Chinese and Korean towns, Mughal and Rajput forts, North African medinas, Persian and Central Asian forts), unless dated by hand as a European work, together with a few known false matches such as Maginot Line blocks. The result is 1,307 forts: 769 with an outline, 498 Wikidata points and 40 points added from Wikipedia.

3.3 Choosing a date

Each fort takes the first available of: the hand date; Wikidata's start time or opening; Wikidata's inception if precise to the year or decade; the OpenStreetMap start date. Dates before 1450 are the

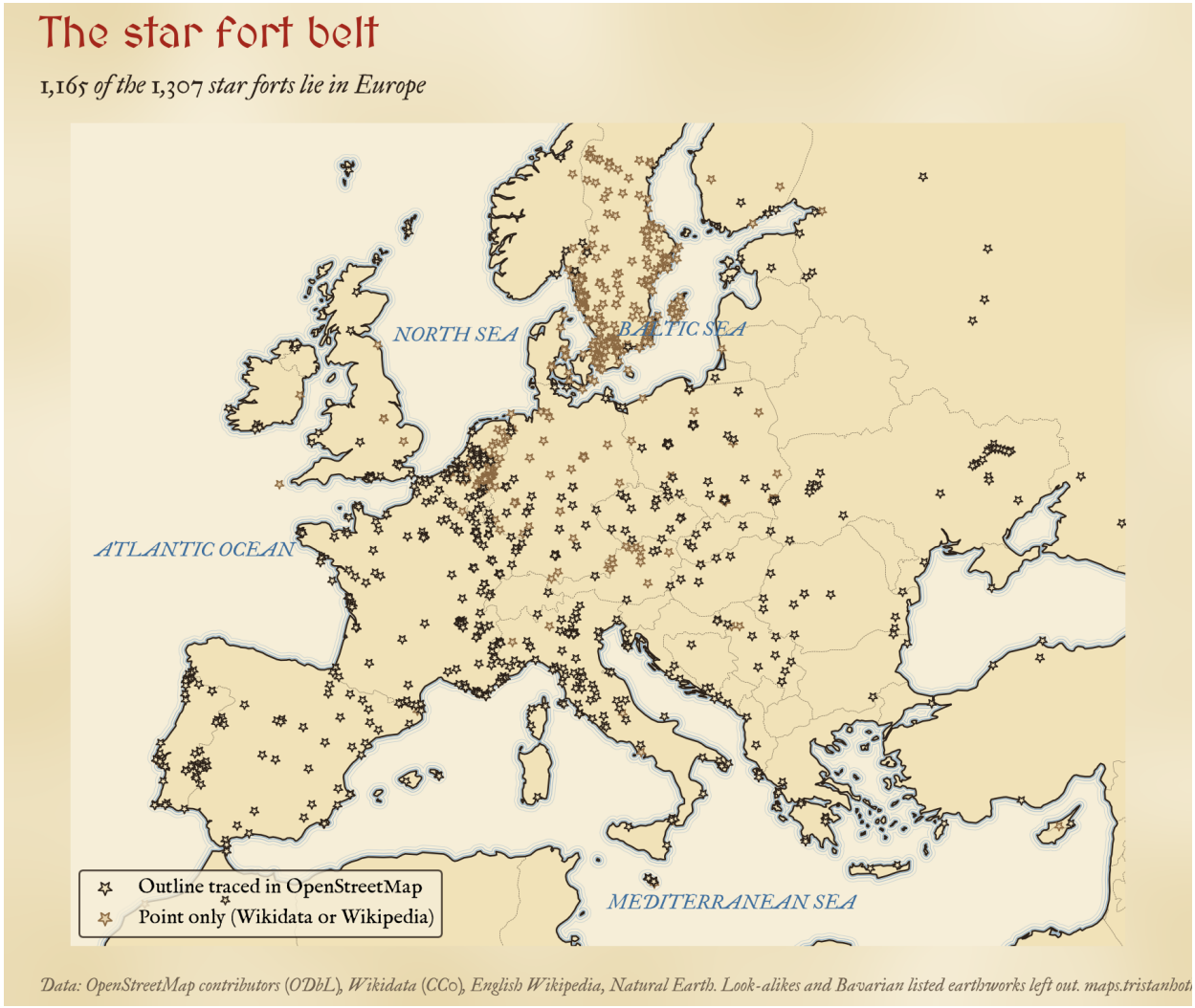


Figure 1: Star forts in Europe and around the Mediterranean, by whether an outline is traced in OpenStreetMap.

medieval origin of a site rather than its bastions and are set aside (10 forts), as are inceptions of monasteries and decade-precision inceptions before 1550. A country's first fort is its earliest dated fort in the data, and its distance is the great-circle distance from Florence to that fort. Diffusion is tested with Spearman's rank correlation between first year and distance.

4 Results

4.1 The belt

1,165 of the 1,307 forts (89%) are in Europe (Figure 1). They form a belt along the early modern war zones: the Low Countries and northern France, the Rhine, the Alps and the Po valley, the Venetian and Habsburg frontiers, the border between Spain and Portugal, and the Baltic. The densest one-degree cell, in the Netherlands around the Rhine and Maas delta, holds 39 forts. Malta has the highest density by far, then the Netherlands (24.9 per 10,000 km^2) and Belgium (9.1). Sweden has the most forts of any country (297), but 295 of them are small sconces catalogued from the national heritage register, a reminder that the catalogue measures recording as well as building (Table 1).

Table 1: The twelve countries with most star forts: all forts, those with a traced outline, and those with a date.

Country	Forts	Outlines	Dated
Sweden	297	2	7
France	170	169	30
Netherlands	93	27	12
Italy	88	83	29
Poland	66	34	14
Germany	60	23	14
Spain	55	54	3
Portugal	47	46	11
United States of America	39	37	18
Ukraine	30	30	3
Belgium	28	13	7
United Kingdom	26	21	12

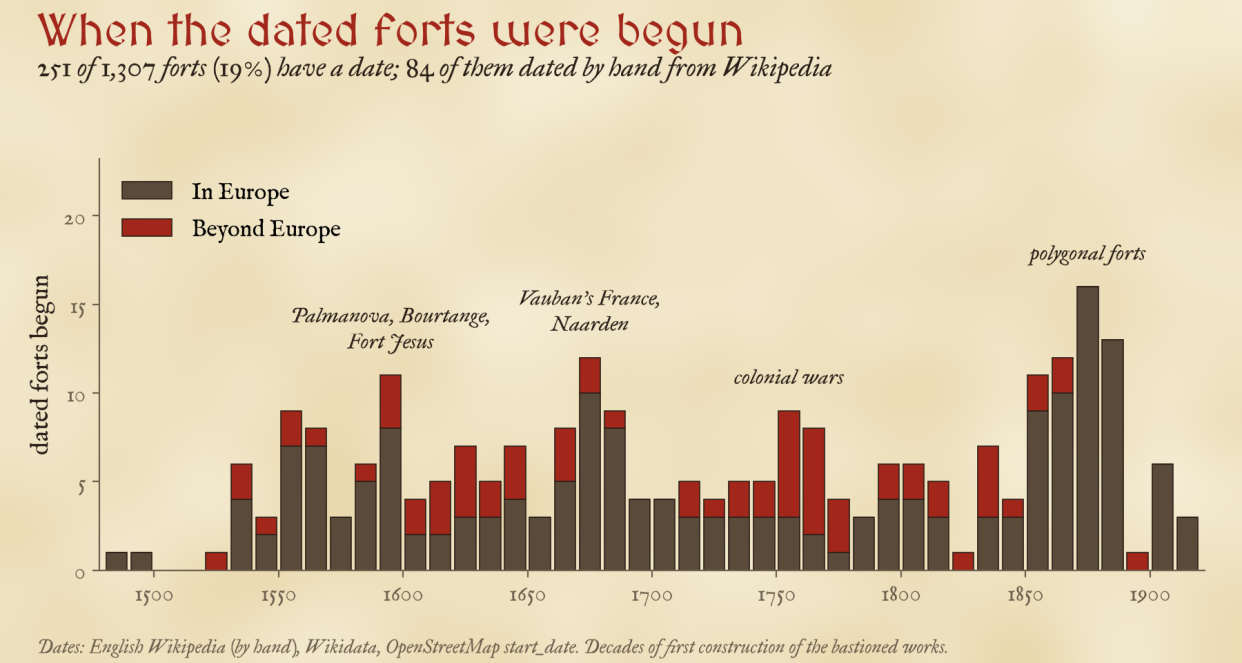


Figure 2: Dated star forts by decade begun, in Europe and beyond.

4.2 Dates

251 forts (19%) have a date: 84 by hand, 127 from Wikidata and 40 from OpenStreetMap. Without the hand dates only 13% would be dated. Of the dated forts, 20% were begun before 1600, 26% in the 1600s, 21% in the 1700s and 34% from 1800, when polygonal forts with low bastions or caponiers continued the type (Figure 2). The dated forts peak in the 1590s, the 1670s and the 1870s; the dated forts beyond Europe peak in the 1750s and 1760s, the decades of the Seven Years' War.

4.3 Diffusion

The earliest dated works in the data are Italian, Cassero della Fortezza (1488) and the Fortezza da Basso in Florence (1534). Table 2 gives the earliest dated fort in each region. The design reached the Low Countries, the German lands, England and Sweden in the 1540s and 1550s, but it reached

Table 2: Regions: forts, dated forts and their share (%), the earliest dated fort in the data and the median year of the dated forts.

Region	Forts	Dated	%	Earliest dated	Year	Median
Italy and Malta	97	34	35	Cassero della Fortezza	1488	1725
Iberia	102	14	14	Fort of São Julião da Barra	1553	1623
France	171	30	18	Fort-Carré	1553	1723
Low Countries	125	19	15	Fort Rammekens	1547	1673
German lands and Central Europe	176	32	18	Citadel of Jülich	1548	1806
Scandinavia and the Baltic	344	21	6	Landskrona Citadel	1559	1677
British Isles	34	14	41	Berwick-upon-Tweed walls	1558	1858
Eastern Europe, Balkans and Turkey	116	17	15	Venetian walls of Nicosia	1567	1734
Africa	15	4	27	Mazagan	1541	1575
Americas	87	42	48	Castillo San Felipe del Morro	1539	1760
Asia and Oceania	40	24	60	Fort Kastela	1522	1644

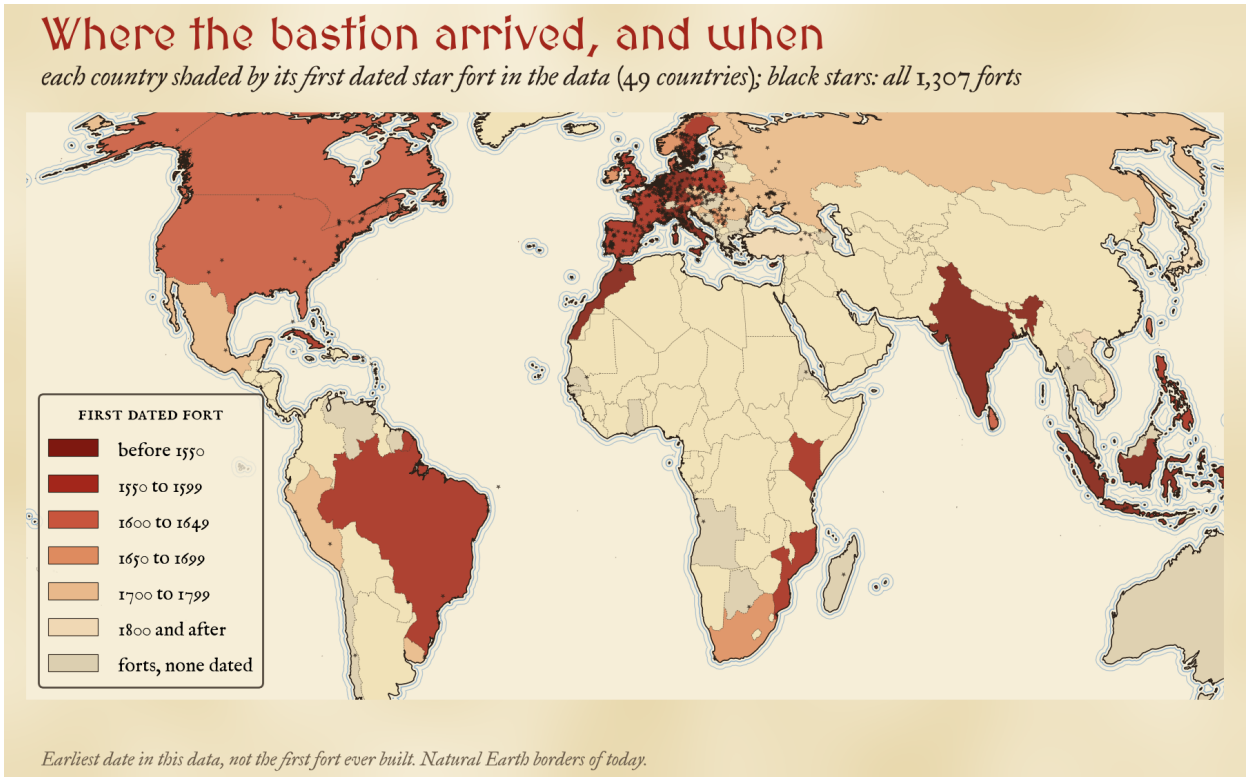


Figure 3: Countries by the period of their first dated star fort in the data. Brown-grey: forts present, none dated.

India, Morocco and Puerto Rico in the same decades (Figure 3). Across the 49 countries with a dated fort, distance from Florence barely predicts the date of the first one ($\rho = 0.15$, $p = 0.32$; within Europe $\rho = 0.29$ across 27 countries, $p = 0.15$). Countries beyond Europe have a median first date of 1623, against 1626 in Europe (Figure 4). The bastion travelled by sea, with the Portuguese and Spanish empires, as fast as it travelled by land.

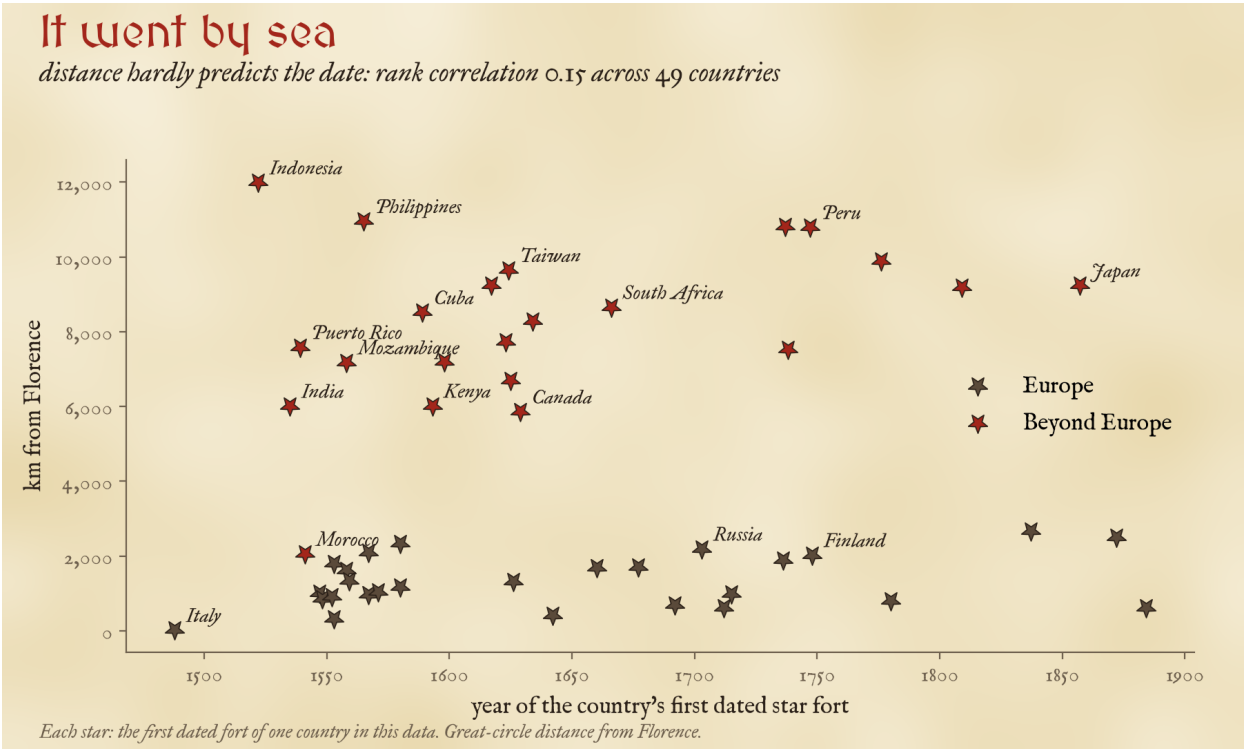


Figure 4: Year of each country's first dated star fort against its distance from Florence.

4.4 The colonial chain

33 of the hand-dated forts are outside Europe (Figure 5). Their order follows the empires: Portugal from Diu (1535), Spain from San Juan (1539), the Dutch East India Company from Banda Neira (1611), the English East India Company from Madras (1640), France on Cape Breton Island (1720), Denmark on St Croix (1738), the United States at Baltimore (1798) and, last, the Tokugawa shogunate at Hakodate (1857). The selection is mine and favours famous forts, so the chain is an illustration rather than a measurement.

4.5 Size and form

The 769 outlines have a median area of 1.9 ha (interquartile range 0.7 to 4.0 ha) and a median diameter of 228 m; 53% enclose less than 2 ha, 89 more than 10 ha and 9 more than 50 ha (Figure 6). Four points is the commonest form (320 outlines, 42%), and 72% have four, five or six. The largest outlines are fortress towns and nineteenth-century ring fortresses (Table 3). Among the dated outlines, size does not change with date (ρ = -0.02, 172 forts): the median is 1.4 ha before 1600, 2.6 ha in the 1600s, 1.6 ha in the 1700s and 2.4 ha from 1800. Figure 7 draws twelve famous outlines at one scale.

4.6 Survival

59% of the forts have an outline that the shape test still recognises; the share is 56% in Europe and 82% beyond it, where the catalogue is made mostly of large, well-known forts. Of the 84 hand-dated forts, 46% have a traced outline; 5 are known to have been largely demolished (the citadels of Antwerp and Turin, Saarlouis, Fort Amsterdam on Manhattan and Louisbourg, the last partly rebuilt). A missing outline is not proof of loss: it can also mean a fort mapped in pieces, or one



Figure 5: Hand-dated star forts outside Europe, in order of the year begun, coloured by builder.

too eroded or too irregular to pass the test.

5 Limitations

First, the catalogue is what has been mapped and catalogued. OpenStreetMap is richest in Europe; Wikidata's sconces come in bulk from a few national registers (Sweden, Bavaria, the Netherlands); and countless earthwork sconces and siege works left no trace. Second, the shape test both misses forts (irregular traces, forts mapped as fragments, coastal forts that use the shore) and finds look-alikes; I removed the look-alikes by country, a blunt rule that also removes some genuine forts. Third, only a fifth of the forts are dated, and the hand-dated ones are a selection of famous forts, so the timeline and the colonial chain lean to the well known. Fourth, inception dates can refer to

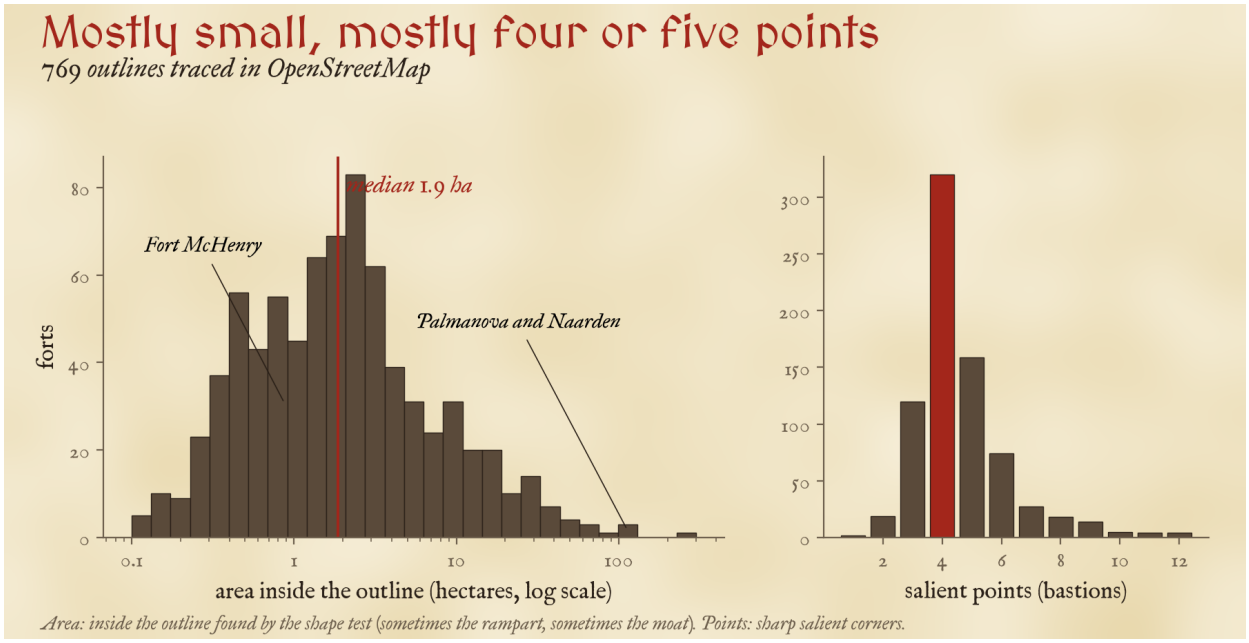


Figure 6: Area inside the traced outlines (log scale) and number of salient points.

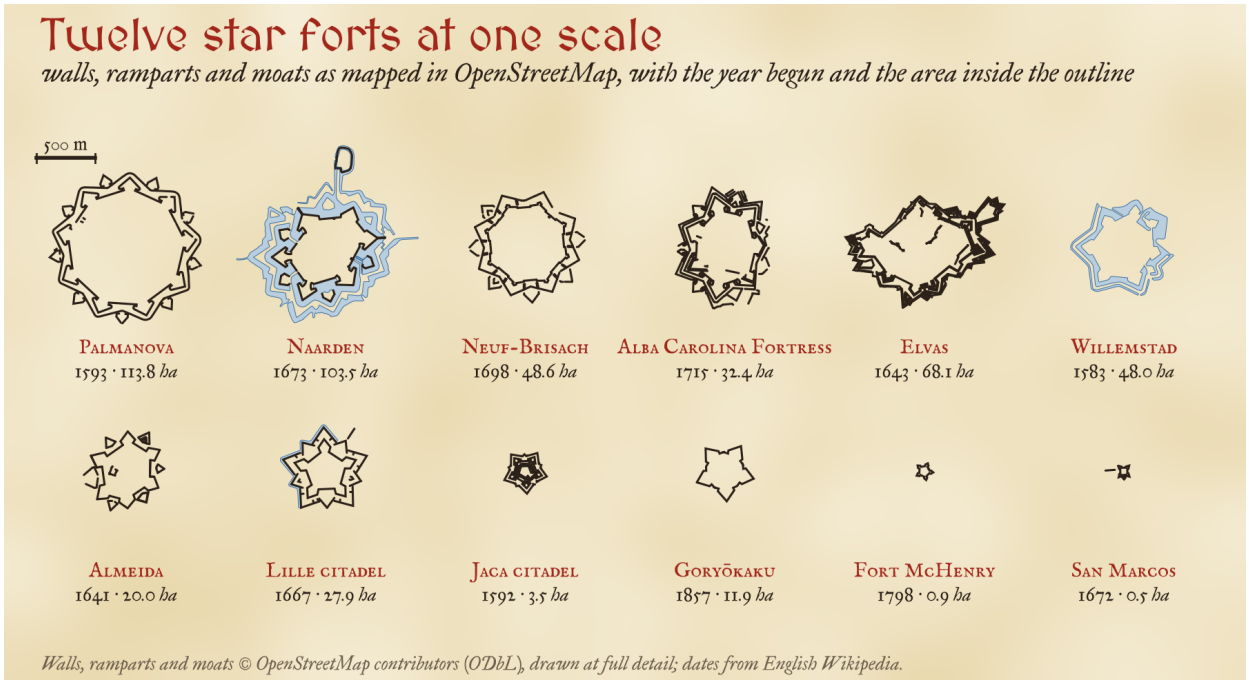


Figure 7: Twelve star forts at one scale, from their OpenStreetMap outlines.

Table 3: The twelve largest traced outlines (unnamed ones carry the name of the nearest town).

Fort	Country	Area (ha)	Points	Begun
Twierdza Modlin	Poland	228.8	5	1806
Star fort near Pisa	Italy	130.7	4	
Palmanova	Italy	113.8	8	1593
Naarden	Netherlands	103.5	5	1673
Fort Monroe National Monument	United States of America	80.0	4	1834
Elvas	Portugal	68.1	17	1643
Valletta	Malta	66.5	12	1566
Famagusta within the walls	Cyprus	59.3	7	
Festung Mont Royal	Germany	51.0	3	
Star fort at Rodos	Greece	49.5	7	
Neuf-Brisach	France	48.6	8	1698
Willemstad	Netherlands	48.0	4	1583

a site's first fortification rather than its bastions. Fifth, the area of a fort is the area inside the line the shape test found, sometimes the inner rampart and sometimes the outer edge of a moat. Sixth, the country of each fort is its modern country.

6 Conclusion

Seen as data, the star fort is a European belt with a long colonial tail. The dated record, thin as it is, shows the design leaving Italy by two routes at once: overland to the war zones of northern Europe, and by sea to the Portuguese and Spanish ports of three continents, where it arrived as early as it did in much of Europe. Most forts were small and square or pentagonal; a few were towns. Two in five no longer show their star to a shape test.

Data and code availability

The interactive map is at https://maps.tristanhotham.com/star-forts/. Data and code are at https://github.com/Tristan-Hotham/star-forts and archived with this paper at https://doi.org/10.5281/zenodo.24043
Tables derived from OpenStreetMap (the outlines and the fort table) are released under the Open Database License 1.0; Wikidata content is CC0; text and figures are CC BY 4.0 and code MIT.

AI statement

I wrote the text, with drafting and editing assistance from Claude (Anthropic), which also helped prepare the analysis code. I reviewed and checked all text and results.

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Wikipedia contributors (2026) English Wikipedia articles on the forts listed in Table 4. CC BY-SA 4.0. https://en.wikipedia.org/

A The hand-dated forts

Table 4: The 84 forts dated by hand from English Wikipedia: year the bastioned works were begun, builder, whether an outline is traced, and fate. Article links are in data/hand_dates.csv.

Fort	Country	Begun	Builder	Outline	Today
Forte Spagnolo	Italy	1534	Spain	no	survives
Fortezza da Basso	Italy	1534	Florence	yes	survives
Diu Fort	India	1535	Portugal	no	survives
Castel Sant'Elmo	Italy	1537	Spain	no	survives
Castillo San Felipe del Morro	Puerto Rico	1539	Spain	no	survives
Mazagan	Morocco	1541	Portugal	no	survives
Citadel of Jülich	Germany	1548	Jülich-Cleves-Berg	no	survives
Fort St Elmo	Malta	1552	Knights of St John	yes	survives
Fort of São Julião da Barra	Portugal	1553	Portugal	no	survives
Rocroi	France	1555	France	no	survives
Berwick-upon-Tweed walls	United Kingdom	1558	England	no	survives
Fort São Sebastião	Mozambique	1558	Portugal	yes	survives
Moti Daman Fort	India	1559	Portugal	no	survives
Spandau Citadel	Germany	1559	Brandenburg	no	survives
Fortezza Medicea di Siena	Italy	1561	Florence	yes	survives
Venetian walls of Bergamo	Italy	1561	Venice	no	survives
Citadel of Turin	Italy	1564	Savoy	no	vanished
Valletta	Malta	1566	Knights of St John	yes	survives
Citadel of Antwerp	Belgium	1567	Spain	no	vanished
Venetian walls of Nicosia	Cyprus	1567	Venice	no	survives
Citadel of Pamplona	Spain	1571	Spain	yes	survives
Zamość	Poland	1580	Poland-Lithuania	no	survives
Willemstad	Netherlands	1583	Dutch Republic	yes	survives
Wülzburg	Germany	1588	Brandenburg-Ansbach	no	survives
Morro Castle	Cuba	1589	Spain	yes	survives
Fort Santiago	Philippines	1590	Spain	yes	survives
Citadel of Jaca	Spain	1592	Spain	yes	survives
Bourtange	Netherlands	1593	Dutch Republic	yes	survives
Fort Jesus	Kenya	1593	Portugal	no	survives
Palmanova	Italy	1593	Venice	yes	survives
Star Castle	United Kingdom	1593	England	no	survives
Forte dos Reis Magos	Brazil	1598	Portugal	no	survives
Citadel of Gozo	Malta	1599	Knights of St John	yes	survives
Fort Belgica	Indonesia	1611	Dutch Republic	yes	survives
Fort Aguada	India	1612	Portugal	no	survives
Fortaleza do Monte	Macau	1617	Portugal	yes	survives
Glückstadt	Germany	1617	Denmark-Norway	no	survives
Fort Zeelandia	Taiwan	1624	Dutch Republic	no	survives
Fort Amsterdam	United States of America	1625	Dutch Republic	no	vanished
Kastellet	Denmark	1626	Denmark-Norway	no	survives
Fort Amsterdam (Curaçao)	Curaçao	1634	Dutch Republic	no	survives
Fort St George	India	1640	England	no	survives
Almeida	Portugal	1641	Portugal	yes	survives
Forte de Santa Luzia	Portugal	1641	Portugal	yes	survives
Elvas	Portugal	1643	Portugal	yes	survives
Galle Fort	Sri Lanka	1649	Dutch Republic	yes	survives
Fredericia	Denmark	1650	Denmark-Norway	no	survives
Fort Saint-Nicolas	France	1660	France	yes	survives
Fredrikstad Old Town	Norway	1663	Denmark-Norway	no	survives
Petersberg Citadel	Germany	1665	Electorate of Mainz	no	survives
Castle of Good Hope	South Africa	1666	Dutch Republic	no	survives
Citadel of Lille	France	1667	France	yes	survives

Fort	Country	Begun	Builder	Outline	Today
Fort Rotterdam	Indonesia	1667	Dutch Republic	yes	survives
Tilbury Fort	United Kingdom	1670	England	yes	survives
Castillo de San Marcos	United States of America	1672	Spain	yes	survives
Naarden	Netherlands	1673	Dutch Republic	yes	survives
Charles Fort	Ireland	1677	England	yes	survives
Mont-Louis	France	1679	France	yes	survives
Jaffna Fort	Sri Lanka	1680	Dutch Republic	yes	survives
Saarlouis	Germany	1680	France	no	vanished
Citadel of Saint-Martin-de-Ré	France	1681	France	yes	survives
Citadel of Blaye	France	1685	France	yes	survives
Petrovaradin Fortress	Serbia	1692	Habsburg Monarchy	no	survives
Neuf-Brisach	France	1698	France	yes	survives
Peter and Paul Fortress	Russia	1703	Russia	no	survives
Fort Marlborough	Indonesia	1714	Great Britain	no	survives
Alba Carolina Fortress	Romania	1715	Habsburg Monarchy	yes	survives
Fortress of Louisbourg	Canada	1720	France	yes	vanished
Fort Christiansvaern	United States Virgin Islands	1738	Denmark-Norway	no	survives
Fortaleza del Real Felipe	Peru	1747	Spain	no	survives
Fort George	United Kingdom	1748	Great Britain	yes	survives
Suomenlinna	Finland	1748	Sweden	no	survives
Castle of Sant Ferran	Spain	1753	Spain	no	survives
Fort Ticonderoga	United States of America	1755	France	no	survives
Fort William	India	1758	Great Britain	no	survives
Fortaleza de Santa Teresa	Uruguay	1762	Portugal	yes	survives
Fortaleza de San Carlos de la Cabaña	Cuba	1763	Spain	yes	survives
Forte de Nossa Senhora da Graça	Portugal	1763	Portugal	yes	survives
Fortaleza de São José de Macapá	Brazil	1764	Portugal	no	survives
Fuerte de San Diego	Mexico	1778	Spain	no	survives
Josefov	Czech Republic	1780	Habsburg Monarchy	no	survives
Terezín	Czech Republic	1780	Habsburg Monarchy	no	survives
Fort McHenry	United States of America	1798	United States	yes	survives
Goryōkaku	Japan	1857	Japan	yes	survives